

Mahim Katiyar

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala	2023 – Present
B.E. in Computer Engineering — CGPA: 8.39/10	
Woodbine Gardenia School, Kanpur	2022
Class 12 (CBSE) — Percentage: 95%	
Woodbine Gardenia School, Kanpur	2020
Class 10 (CBSE) — Percentage: 97.8%	

SKILLS

Languages: C, C++, Python, SQL
Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems (OS), Computer Networks (CN)
Databases: MySQL, Oracle
Backend: FastAPI, Pydantic, REST APIs
AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP)
Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG)
Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn
Libraries/Tools: Pandas, NumPy, Matplotlib, NLTK, spaCy, Hugging Face Transformers, Docker, Git, GitHub
Spoken: English (Fluent), Hindi (Fluent)

PROJECTS

Pneumonia Detection using Transfer Learning GitHub | Demo

Tech: TensorFlow, PyTorch, Keras, DenseNet121, ResNet18, MobileNetV2

- Built a binary classification pipeline to detect pneumonia from chest X-rays using transfer learning, benchmarking DenseNet121, ResNet18, and MobileNetV2 across TensorFlow/Keras and PyTorch.
- Fine-tuned ImageNet-pretrained backbones in two phases with custom image preprocessing, reaching **90.87% accuracy**.
- Evaluated models on accuracy, ROC-AUC, precision, recall, and F1-score to select the best-performing architecture.

Python Package for TOPSIS (Multi-Criteria Decision Making) GitHub | Demo

Tech: Python, NumPy, Pandas, CLI packaging

- Built and packaged a reusable Python library implementing the TOPSIS algorithm, following standard conventions for PyPI-style distribution.
- Designed a CLI supporting CSV/XLSX inputs with automated validation of weights and impacts.
- Implemented weighted normalization, ideal-solution computation, and ranking logic for multi-criteria decision analysis.

Resume Analyzer – AI-Powered Resume Screening Tool GitHub | Demo

Tech: FastAPI, Streamlit, spaCy, Sentence-Transformers, Docker

- Built a full-stack NLP web app that parses resumes (PDF/DOCX) and extracts skills and named entities using spaCy PhraseMatcher and NER, served through a documented FastAPI REST API and a Streamlit UI.
- Implemented semantic job-description matching with sentence-transformer embeddings and cosine similarity to score resume-JD fit (0–100) and surface missing skills, capturing meaning rather than keyword overlap.

EXPERIENCE

ELC Summer Internship, Thapar Institute of Engineering and Technology GitHub | June 2025 – July 2025

- Built an ML/DL pipeline to classify ADHD vs. healthy subjects from a 19-channel EEG dataset (121 subjects, 2.16M records), including preprocessing, stratified train-test split, and leakage-free feature scaling.
- Benchmarked Logistic Regression, Random Forest, and a hybrid LSTM+CNN model combining spatial and temporal EEG feature extraction.
- Achieved best results with the LSTM+CNN model (76.9% accuracy, 0.84 ROC-AUC, 0.85 ADHD recall), outperforming baseline classifiers on non-linear EEG patterns.

POSITIONS OF RESPONSIBILITY

Mess Secretary Aug 2024 – May 2025

- Served as Mess Secretary of Hostel B, overseeing daily mess operations and coordinating with kitchen and service staff to maintain consistent food quality and service standards.
- Led a 30-member team in planning and executing Hostel Night, overseeing coordination and on-ground execution.